

Concept refresher

- **Broadcast domain:** every client that sees the same ARP or DHCP flood. Separate domains, and a worm can't hop in one packet.
- **Access port:** untagged, belongs to one VLAN.
- **Trunk port:** tagged, carries many VLANs between switches and routers.



Five-minute lab – Create three core VLANs

The demo uses a Cisco RV-router GUI, but the logic is identical on MikroTik, Juniper or Fortinet.

VLAN	Purpose	Tag	Example IP Pool
10	Corp-LAN (HR, Finance)	10	192.168.10.0/24
20	Ops-LAN (Dev, IoT)	20	192.168.20.0/24
99	Quarantine	99	192.168.99.0/24

1. **Navigate to LAN > VLAN Settings → Add.** Enter 10 and label “Corp-LAN”. Save. Repeat for 20 and 99.
2. **Assign Access Ports.** Plug HR PCs into switch ports 1-8, set PVID 10. Dev gear on ports 9-12 gets PVID 20. Leave port 24 as trunk (all VLANs tagged) uplinking to the router.
3. **Enable Inter-VLAN Routing Only Where Needed.** In Routing > Static, add Corp-LAN → Internet but deny Corp ↔ Ops. Finance can now surf but not scan test servers.
4. **Test Isolation.** From HR PC run ping [192.168.20.10](#) (should fail). [ping 8.8.8.8](#) (should succeed).
Quick Win: Block UDP/137-138 on Corp-LAN to kill legacy NetBIOS chat; most worms still probe it.

Jump-box isolation for admin tasks

Admins need root on everything, but attackers love RDP. The answer is a hardened “jump box” reachable only from an Admin VLAN and never from a public net. F5 Labs lists three must-do steps: isolate, restrict ingress, monitor.

Step-by-step

1. **Create VLAN 30 – Admin.** No DHCP; require static IPs to deter rogue plugs.
2. **Deploy a minimal VM (Ubuntu LTS, two NICs).** One NIC on Admin VLAN, the other in Quarantine to patch.
3. **Lock RDP/SSH to Jump Box.** Edge firewall ACL: src Admin-VLAN dest Any-Internal allow RDP/SSH; all others deny.
4. **MFA Gate.** Install PAM-Google Authenticator or Duo. Now even stolen passwords fail.
5. **Session Logging.** Syslog every command to your SIEM so mistakes are traceable.

Result: Even if an admin laptop is phished on home Wi-Fi, the attacker can't leapfrog into servers without clearing the MFA wall.

Micro-segmentation lite

Full SDN micro-segmentation costs lakhs. VLAN + ACL + Jump-Box gets 80% of the benefit for one-tenth the spend, according to Cisco's white paper on small-office Zero-Trust.

2) QoS tuning so video calls don't choke payroll traffic

The bandwidth myth

“We have 300 Mbps fibre – why do Teams calls still freeze?” Because bandwidth isn't the bottleneck; bufferbloat and queue starvation are. Fortinet's traffic-shaping guide says QoS only matters “when links saturate – then priority wins.”



Decide what deserves priority

1. **Platinum:** VoIP, video calls (EF DSCP 46).
2. **Gold:** SaaS payroll, banking portals.
3. **Silver:** Git pulls, cloud backups.
4. **Best-Effort:** Social media, YouTube.
5. **Scavenger:** Torrent, OS updates.

Configure on a FortiGate (FortiOS 7.6)

A. Create traffic classes

```
config firewall shaper traffic-shaper
```